

Module 05: CSMA with Collision Avoidance (CSMA/CA) & Wireless LANs

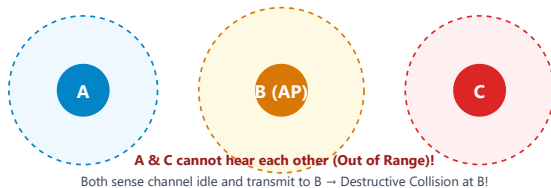
Unit 2: Data Link Layer & MAC Sublayer

IEEE 802.11 Standards

In-depth architectural analysis of collision avoidance in wireless networks, solving the Hidden and Exposed Station problems via SIFS/DIFS timers, RTS/CTS handshake, and NAV virtual carrier sensing.

CSMA/CA Architecture: Collision Avoidance, IFS & RTS/CTS Handshake (IEEE 802.11)

1. Why CSMA/CD Fails in Wireless (Hidden Stations)

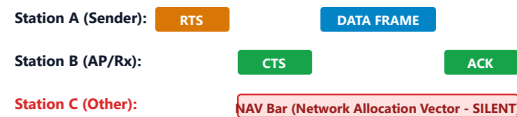


• Hardware Limitation in RF Transceivers:

Station's own transmission power drowns out received signals (Dynamic Range).
Collision detection during transmission is impossible in standard Wi-Fi.

Solution: Avoid collisions beforehand using IFS & Virtual Sensing!

2. RTS / CTS Virtual Sensing with NAV



• The 4-Way Handshake Steps:

1. A sends RTS (contains duration of entire upcoming transaction).
2. B responds with CTS (echoes remaining duration to all neighbors including C).
3. C hears CTS → Sets NAV timer → Suppresses transmission until ACK completes.
4. A transmits Data safely without any risk of collision from Station C!

• **SIFS < PIFS < DIFS ensures highest priority for control packets (ACK, CTS).**

SIFS (Shortest) = CTS / ACK | DIFS (Distributed) = New Data Contention

CSMA/CA Protocol Mechanics & Interframe Space (IFS) Hierarchy

IFS Hierarchy: SIFS (Shortest) < PIFS (PCF Mode) < DIFS (DCF Mode) < EIFS (Extended / Error)

Contention Window: Station senses idle channel → Waits DIFS → Chooses random Backoff from [0, CW]. CW doubles on collision: $CW_{min} = 15$, $CW_{max} = 1023$.

Exposed Station: Station B transmits to A. Station C wants to transmit to D (away from B). Senses B and unnecessarily halts. RTS/CTS solves this cleanly.

NAV Mechanism: Virtual Carrier Sensing: Stations do not waste battery continuously sniffing RF; hardware timer counts down NAV to zero before sensing.

1. Why CSMA/CD Fails in Wireless Networks

In wired Ethernet, collision detection is performed by sensing voltage spikes ($V_{channel} > 2 \cdot V_{trans}$). In wireless media, radio signal power attenuates rapidly with distance ($1/r^2$). Due to transceiver dynamic range limitations, a station's powerful transmitter blinds its sensitive receiver amplifier, making real-time collision detection virtually impossible.

2. Hidden and Exposed Station Dilemma

- **Hidden Station Problem:** Stations A and C cannot detect each other's RF signals due to physical distance or obstacles. Both sense the channel as idle and transmit to intermediate Access Point B simultaneously, resulting in catastrophic collision at station B.
- **Exposed Station Problem:** Station C unnecessarily refrains from transmitting to D because it senses active transmission between A and B, despite D being far outside B's interference range.

3. 4-Way RTS/CTS Handshake & NAV Virtual Sensing

NAV Duration Equation

$$\text{Duration_RTS} = \text{SIFS} + \text{T_CTS} + \text{SIFS} + \text{T_DATA} + \text{SIFS} + \text{T_ACK}$$

$$\text{NAV_CTS} = \text{SIFS} + \text{T_DATA} + \text{SIFS} + \text{T_ACK}$$

Every station overhearing a CTS sets its **Network Allocation Vector (NAV)** countdown timer to the duration value, silencing transmissions until the complete data-ACK cycle finishes.